

Section 1: Executive Summary

The Grand Marina's Hotel Relies on The HYDROLOGIC management system which monitors the water flow across the hotel's three water service lines , The HYDROLOGIC Devices monitors The water Service Line in which it continuously sends Vital Gate positions,Water consumption Data and communications between the devices and the dashboard. Hotel operators and management can monitor water consumption and remotely adjust gates or activate an emergency shutoff.

The threat model is important because this system connects the hotel's digital system to the physical water controls.A cybersecurity incident could therefore affect more than just computers or data: it could disrupt the water services,cause incorrect water flow settings,or prevent hotel staff from responding to an emergency.

The assessment identified three priority areas:

1. Unauthorized access to the system and controls
2. Disruption of communication between the devices and cloud services
3. Unauthorized access (to /manipulation of) Water consumption information.

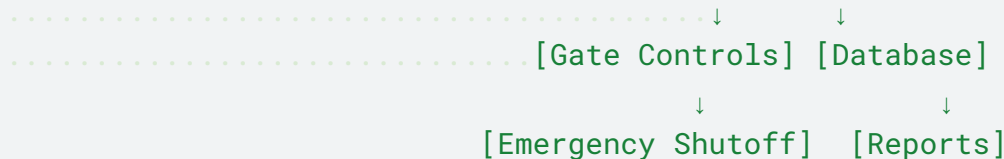
Section 2: System Overview

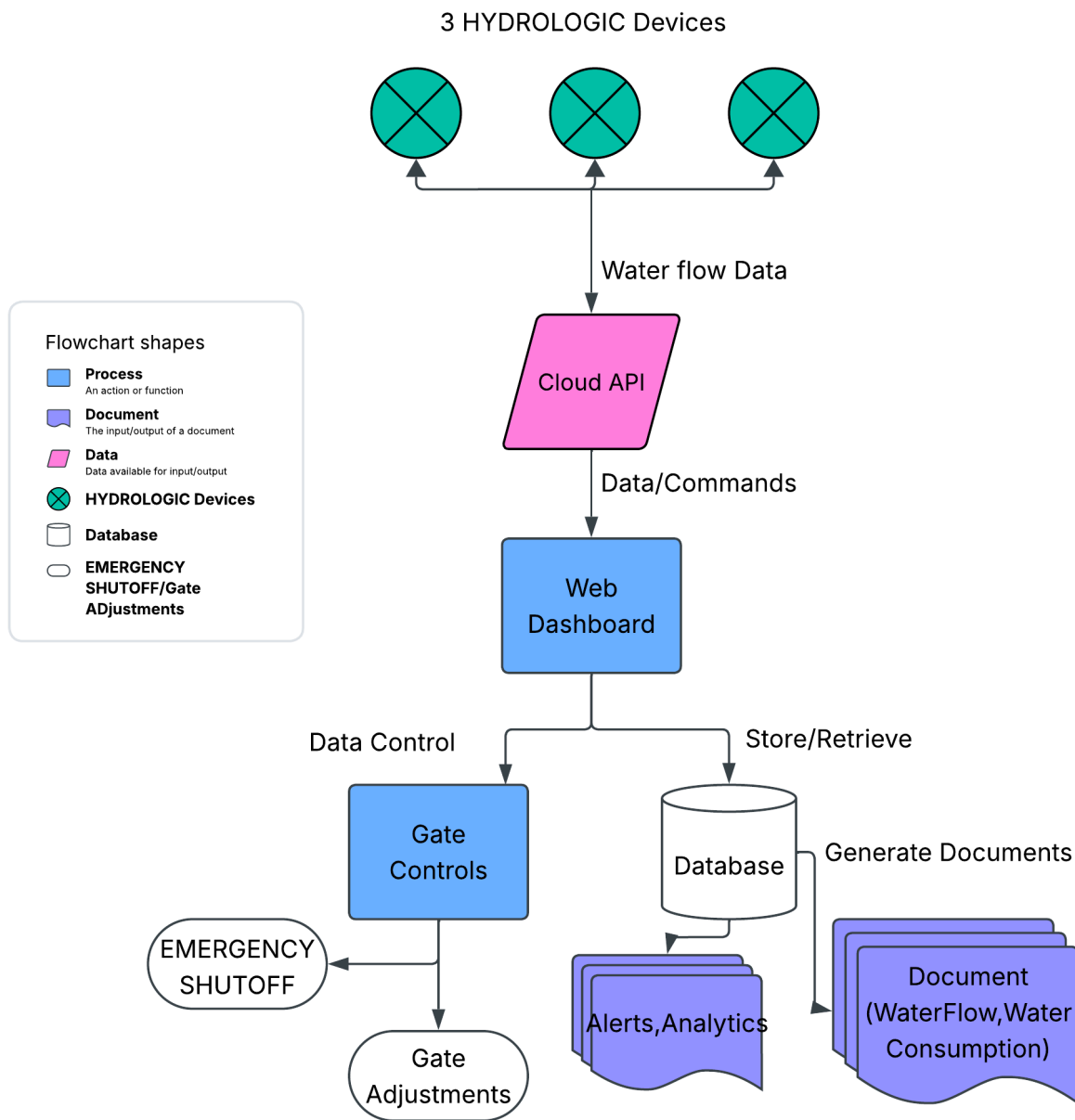
The Grand Marina is a 500-room luxury hotel using Hydroficient's HYDROLOGIC system to monitor and manage its water consumption.The hotel Has three HYDROLOGIC management devices,with one installed on each water service line(Main Building, Pool/Spa, Kitchen/Laundry). These devices obtain water flow vital information and communicate with hydroficient's cloud based services.

Hotel operators and management can access the system through a web-based dashboard that allows authorized personnel to monitor water usage,reports,and control water system functions such as adjusting gates and activating an emergency water shutoff.

None

[HYDROLOGIC Devices] → [Cloud API] → [Dashboard]





How it works:

- The 3 HYDROLOGIC devices sends vitals every few seconds
- Data Transmits via Cloud API and to the dashboard
- Cloud processes the data
- Dashboard displays real time information to the operator
- Via the dashboard, Actions can be taken such as: emergency shutoff, Gate adjustments, Viewing the Waterflow, Water consumptions, etc...

Section 3: Asset Inventory

List the key assets and their CIA priorities (use your work from Step 2):

Asset	Description	C Priority	I Priority	A Priority
HYDROLOGIC Devices	3 flow management units	High	High	High
Web Dashboard	Operator monitoring interface	High	High	High
Cloud API	Device-to-cloud communication	High	Critical	Critical
Remote Controls	Gate adjustments, emergency shutoff	High	Critical	Critical
Consumption Data	Savings reports, billing records	High	High	Medium

Priority Rationale

- **Integrity is Critical** for remote controls and the Cloud API — unauthorized changes could cause incorrect water-flow settings or an unwanted emergency shutoff.
- **Availability is Critical** for HYDROLOGIC devices, the Cloud API, and remote controls — downtime could prevent the hotel from monitoring or managing its water system.
- **Confidentiality is High** for consumption data — unauthorized access could expose the hotel's water usage, savings, and billing-related information

Section 4: STRIDE Analysis

This is the core of your threat model. For **each major component**, analyze all six STRIDE categories:

Component 1: HYDROLOGIC Devices

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	An attacker impersonates a legitimate Hydrologic device and attempts to communicate with cloud API	Medium	Critical	High
Tampering	An attacker modifies commands or configuration data sent to a device causing incorrect waterflow settings	Medium	Critical	High
Repudiation	An unauthorized command is sent through a compromised device,insufficient logging makes it hard to determine which device initiated it.	High	Medium	Medium
Info Disclosure	An attacker gains access to data transmitted by a device and obtains information about the	Medium	Medium	Medium

	hotel's consumption or operation.			
Denial of Service	An attacker disrupts a device's network connection, preventing it from communicating with the cloud and being monitored remotely	Medium	High	High
Elevation of Privilege	An attacker exploits a device-management weakness to obtain privileges reserved for authorized personnel	Low	High	High

Component 2: Web Dashboard

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	An attacker obtains an operator's credentials and logs into the dashboard while pretending to be that employee	Medium	Critical	High
Tampering	An attacker with unauthorized dashboard access changes waterflow settings or sends commands	Medium	Critical	Critical
Repudiation	An authorized or compromised account performs a dangerous action, but inadequate audit logs makes it difficult to determine who performed it	Medium	High	High
Info Disclosure	An unauthorized user access the dashboard and views water consumption information	Medium	Medium	Medium

Denial of Service	An attacker prevents operators from accessing the dashboard	Medium	High	High
Elevation of Privilege	A dashboard user exploits an authorization vulnerability to obtain administrator privileges and access remote control functions.	Low	Critical	High

Component 3: Cloud API

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	An attacker impersonates a legitimate Hydrologic device or dashboard and sends request to the cloud API	Medium	Critical	High
Tampering	An attacker intercepts or modifies API requests before reaching the system(changing water controls,	Medium	Critical	High

	commands, and data)			
Repudiation	An attacker sends unauthorized requests,insufficient API logging makes it difficult to trace the requests back to their source.	Medium	High	High
Info Disclosure	An attacker gains unauthorized access to API responses or communications and obtains operational information	Medium	High	High
Denial of Service	An attacker sends excessive requests to the API,preventing legitimate devices and dashboards users from communicating	High	Critical	High
Elevation of Privilege	An attacker exploits API authorization weakness to access functions intended only for administrators	Low	Critical	High

	or trusted devices.			
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Component 4: Remote Controls (Gate/Shutoff)

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	An attacker impersonates an authorized operator and sends a remote gate or shutoff command	Medium	Critical	High
Tampering	An attacker modifies legitimate remote-control so that gate is adjusted incorrectly or emergency shutoff is activated without authorization	Medium	Critical	Critical
Repudiation	Someone activates a remote shutoff, inadequate logging prevents the hotel from determining	Medium	High	High

	who initiated the action			
Info Disclosure	An attacker gains access to information about the state of gates or shutoff controls	Low	Medium	High
Denial of Service	An attacker prevents legitimate remote control commands reaching the devices	Medium	Critical	High
Elevation of Privilege	A user with limited permission exploits an authorized weakness to obtain access to emergency shutoff/gate functions	Low	Critical	High

Section 5: Risk Summary

List your findings by risk level:

Critical Risks

- **Unauthorized control of water gates/shutoff** – Could disrupt the hotel's water supply.
- **Cloud API tampering** – Could allow attackers to modify commands sent to the water-management system.

High Risks

- **Compromised dashboard accounts** — Attackers could gain access to system controls.
- **Cloud API/device denial of service** — Could prevent monitoring and remote control.
- **Privilege escalation** — A low-level user could gain administrative access.
- **Unauthorized access to water-consumption data** — Could expose sensitive operational information.
- **Insufficient logging** — Could make it difficult to identify who performed unauthorized actions.

Medium Risks

- **Device information disclosure** — Could expose system or water-usage information.
- **Insufficient device activity tracking** — Could make investigations more difficult.

Section 6: Recommended Mitigations

For each Critical and high-risk, propose a defense:

Risk	Proposed Mitigation	Implementation Complexity
Example: Weak dashboard authentication	Enable multi-factor authentication	Low — configuration change
Unauthorized control of water gates/shutoff	Enable MFA and restrict remote-control access to authorized staff only	Low
Cloud API tampering	Use encrypted communication and strong API authentication	Medium
Compromised dashboard accounts	Use MFA, strong passwords, and account lockout after repeated failed logins	Low
Cloud API/DDOS	Add rate limiting, monitoring, and backup communication methods	Medium

Privilege escalation	Apply role-based access control and give users only the permissions they need	Medium
Unauthorized access to water-consumption data	Encrypt sensitive data and restrict access based on user roles	Medium
Insufficient logging	Maintain detailed audit logs for logins,configuration changes,and remote control commands	Low